Week 16 — Extended Reading: World models and planning

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Chua et al. (2018): PETS

Primary source. https://arxiv.org/abs/1805.12114

Mechanism and scope. PETS combines probabilistic dynamics ensembles with trajectory sampling and model-predictive control. An ensemble can represent disagreement between fitted models, while each model can also represent transition noise.

Read, derive, investigate. Distinguish epistemic uncertainty from aleatoric noise. Why can an ensemble agree confidently outside the training distribution? Compare planning with one sampled model per trajectory and resampling models along a trajectory; inspect which uncertainty each construction propagates.

2 Janner et al. (2019): MBPO

Primary source. https://arxiv.org/abs/1906.08253

Mechanism and scope. MBPO branches short model rollouts from real data to supplement policy training. Short rollouts address accumulation of model error; they do not make a biased model exact or justify arbitrary synthetic-data ratios.

Read, derive, investigate. Write a one-step model error and show how it can affect subsequent visited states. Explain why starting from a real state changes the error exposure. Design a horizon sweep that keeps real interaction count fixed and also reports model fitting error.

3 Hafner et al. (2023): DreamerV3

Primary source. https://arxiv.org/abs/2301.04104

Mechanism and scope. DreamerV3 learns a latent world model and trains behaviour from imagined trajectories, with a collection of architectural and optimisation choices for scale and stability. Distinguish state inference, latent transitions, reward and continuation prediction from actor/critic training.

Read, derive, investigate. Follow one observation into the posterior latent state and one imagined action into the next latent state. Which losses train reconstruction, reward and continuation? Our tiny latent model uses a deliberately simplified actor estimator; it is not a full DreamerV3 reproduction or a statement about all Dreamer actor-gradient variants.

4 Schrittwieser et al. (2020): MuZero

Primary source. https://arxiv.org/abs/1911.08265

Mechanism and scope. MuZero learns representation, dynamics and prediction functions supporting search. The recurrent dynamics predicts both a next hidden state and reward; prediction supplies policy and value. A pixel-perfect decoder is not required for its planning objective.

Read, derive, investigate. Draw the initial inference and recurrent inference interfaces, including the reward output. Which training targets come from rewards, return bootstraps and search visits? The course world-model project does not implement MuZero tree search; use the diagram to compare algorithms rather than renaming the toy.

5 Alonso et al. (2024): DIAMOND

Primary source. https://arxiv.org/abs/2405.12399

Mechanism and scope. DIAMOND investigates reinforcement learning with a diffusion world model. Its generated visual futures and the policy trained inside them should be assessed both as predictions and as a basis for decision making.

Read, derive, investigate. Why can a visually convincing rollout still misrepresent the consequences of an action? Propose an action-conditioned consistency test. Separate a reported benchmark result from the budget and architecture of this week's small model; no frontier benchmark is reproduced here.

Connect the reading to your workbook

For imagined transition $z_t, a_t \to z_{t+1}$, declare whether the predicted reward and continuation are indexed by departure or arrival. Compute the final bootstrap before recursing backward through the lambda return. Set continuation to zero on a terminal transition and confirm that later values vanish. This numerical contract is more informative than copying a model name onto a different architecture.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Shared model bias can hide behind ensemble agreement. Read bias and regression validation before interpreting world-model agreement. Calculate accumulated discounted error separately from reconstruction loss; retain Dreamer/MuZero primary sources for their particular training objectives and representations.

Checked textbook locations: Siegrist § § 4.1, 7.1, PDF pp.439, 896; Ross § § 9.2–9.4, PDF pp.421–429. https://stats.libretexts.org/@go/page/10189. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.